

Question ID: 6ba39da5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

$$2a + 8b = 198$$
$$2a + 4b = 98$$

The solution to the given system of equations is (a, b) . What is the value of b ?

Correct Answer: 25

Rationale

The correct answer is **25**. Subtracting the second equation from the first equation in the given system of equations yields $(2a - 2a) + (8b - 4b) = 198 - 98$, which is equivalent to $0 + 4b = 100$, or $4b = 100$. Dividing each side of this equation by **4** yields $b = 25$.